

A feedback control approach to convex optimization with inequality constraints

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Outline

- ➊ Introduction and background
 - Primal-dual gradient dynamics
 - Controlled multiplier optimization approach
- ➋ PI-CMO for inequality constrained problems
- ➌ Numerical Examples
- ➍ Conclusion

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Introduction

- The goal of this work is to develop a new algorithm for **inequality constrained**, convex optimization problems:

$$\begin{aligned} \min_{x \in \mathbb{R}^n} f(x) \\ \text{s.t. } h(x) = Cx - d \leq 0 \end{aligned} \tag{1}$$

where $C \in \mathbb{R}^{m,n}$, $d \in \mathbb{R}^m$ and “ $\leq$ ” denotes the componentwise inequality;

- We rely on a **continuous-time** formulation;
- We take a **control** approach to **design** a novel optimization algorithm.

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Augmented primal-dual gradient dynamics

We consider the **augmented primal-dual gradient dynamics (Aug-PDGD)** (see, e.g., [Qu and Li, 2019])

$$\begin{aligned}\dot{x}(t) &= -\nabla_x \mathcal{L}_A(x, \lambda) \\ \dot{\lambda}(t) &= \eta \nabla_\lambda \mathcal{L}_A(x, \lambda), \quad \eta \in \mathbb{R}_{>0}\end{aligned}\tag{2}$$

where

$$\mathcal{L}_A(x, \lambda) = f(x) + \sum_{j=1}^m g_j(x, \lambda_j), \quad g_j(x, \lambda_j) = \begin{cases} \lambda_j h_j(x) + \frac{\rho}{2} h_j^2(x) & \text{if } h_j(x) \geq -\frac{\lambda_j}{\rho}; \\ -\frac{1}{2\rho} \lambda_j^2 & \text{otherwise} \end{cases}\tag{3}$$

is the augmented Lagrangian.

Augmented primal-dual gradient dynamics

According to Theorem 2 in [Qu and Li, 2019]:

Theorem (Global exponential convergence of Aug-PDGD)

The global optima of (1) is a globally exponentially stable equilibrium point of Aug-PDGD, i.e., for any $\eta > 0, \rho > 0$, there exists constant $\tau, C_3, C_4 > 0$ such that

$$\|x(t) - x^*\|_2 \leq C_3 e^{-\frac{1}{2}\tau t}, \quad \|\lambda(t) - \lambda^*\|_2 \leq C_4 e^{-\frac{1}{2}\tau t} \quad (4)$$

Remark 1: The proof is rather technical because it needs a **non-diagonal quadratic Lyapunov function**.

Remark 2: It is difficult to explicitly assess τ from the formula given in Theorem 2 in [Qu and Li, 2019], because it depends on several constants.

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Controlled Multiplier Optimization (CMO)

We recall the CMO approach [Cerone et al., 2024]. For equality-constrained problems

$$\min_x f(x) \quad \text{s.t. } h(x) = 0, \quad (5)$$

we define the dynamic system

$$\mathcal{P}_{eq} : \begin{cases} \dot{x}(t) = -\nabla_x f(x(t)) - \sum_{i=1}^m \nabla_x h_i(x(t)) \lambda_i(t) \\ y(t) = h(x(t)) \end{cases} \quad (6)$$

Lemma (CMO for equality constrained problems)

An equilibrium point (x^, λ^*) of $\mathcal{P}_{eq}$ is a stationary point of problem (5) iff $h(x^*) = 0$.*

The optimization problem is equivalent to a stabilization and regulation control problem.

PI-CMO for equality constrained problems

Design a **PI controller** for $\mathcal{P}_{eq}$. The corresponding closed-loop dynamics is the PI-CMO dynamics:

$$\begin{aligned}\dot{x} &= -\nabla f(x) - J_h(x)^\top \lambda \\ \dot{\lambda} &= -K_p J_h(x) \left[\nabla f(x) + J_h(x)^\top \lambda \right] + K_i h(x).\end{aligned}\tag{7}$$

- If $K_p = 0$, we get the standard PDGD per equality-constrained problems; i.e., PDGD corresponds to integral control;
- PI-CMO is globally exponentially stable (see next slide);
- We can show that there is always a K_p such that PI-CMO has a faster convergence rate than PDGD.

Theorem (Global exponential convergence of PI control method)

Let assumptions $h = Cx + d$ with D full rank. For any $K_p > 0$, given a constant $\rho \geq \frac{K_p(\beta_2 - \beta_1)^2}{2\beta_1}$, let us set either $K_i = \rho + K_p\beta_2$ or $K_i = \rho + K_p\beta_1$. Then, there exist real positive constants c_1 and c_2 such that, for the dynamics (7),

$$\|x(t) - x^*\|_2 \leq c_1 e^{-\frac{1}{2}\tau t}, \quad \|\lambda(t) - \lambda^*\|_2 \leq c_2 e^{-\frac{1}{2}\tau t} \quad (8)$$

where

$$\tau = \min \left\{ K_p \alpha_1, 2\beta_1 - \frac{K_p(\beta_2 - \beta_1)^2}{\rho} \right\}. \quad (9)$$

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KKT conditions

Theorem (KKT first-order conditions)

Let $x^* \in \mathbb{R}^n$ be a minimum of f subject to the constraints $h(x) \leq 0$. Then, there exists a unique $\lambda^* \in \mathbb{R}^m$ such that

$$\begin{aligned}\nabla f(x^*) + J_h(x^*)^\top \lambda^* &= 0 \\ \lambda_i^* h_i(x^*) &= 0, \quad i = 1, \dots, m \\ \lambda^* &\geq 0, \quad h(x^*) \leq 0.\end{aligned}\tag{10}$$

CMO lemma for inequalities

Let $\mathcal{L}_A$ be the augmented Lagrangian (3). Define the dynamic system

$$\mathcal{P} : \begin{cases} \dot{x}(t) = -\nabla_x \mathcal{L}_A(x(t), \lambda(t)) \\ y(t) = -\nabla_\lambda \mathcal{L}_A(x(t), \lambda(t)) \end{cases} \quad (11)$$

Lemma (CMO for inequality constrained problems)

An equilibrium point (x^, λ^*) of $\mathcal{P}$ satisfies the KKT conditions (10) for optimization problem (1) if and only if $\nabla_\lambda \mathcal{L}_A(x^*, \lambda^*) = 0$.*

The PI controller

Closing the loop on $\mathcal{P}$ using a PI controller gives:

$$\begin{aligned}\dot{x} &= -\nabla_x \mathcal{L}_A(x, \lambda) \\ \dot{\lambda} &= K_i \nabla_\lambda \mathcal{L}_A(x, \lambda) + K_p \frac{d}{dt} \nabla_\lambda \mathcal{L}_A(x, \lambda).\end{aligned}\tag{12}$$

The PI controller

Closing the loop on $\mathcal{P}$ using a PI controller gives:

$$\begin{aligned}\dot{x} &= -\nabla_x \mathcal{L}_A(x, \lambda) \\ \dot{\lambda} &= K_i \nabla_\lambda \mathcal{L}_A(x, \lambda) + K_p \frac{d}{dt} \nabla_\lambda \mathcal{L}_A(x, \lambda).\end{aligned}\tag{12}$$

↑
Discontinuous!

- The discontinuity of the differential equation describing the closed-loop system creates possible issues. In particular, the solution flow may be non-unique.

A modified augmented PI-CMO dynamics

We modify (12) as follows:

$$\begin{aligned}\dot{x} &= -\nabla_x \mathcal{L}_A(x, \lambda) \\ \dot{\lambda} &= K_i \nabla_\lambda \mathcal{L}_A(x, \lambda) + K_p J_h(x) \dot{x}\end{aligned}\tag{13}$$

- Basically, we replace $\frac{d}{dt} \nabla_\lambda \mathcal{L}_A(x, \lambda)$ by $J_h(x) \dot{x}$ which may be interpreted as taking a continuous approximation of (12).
- We refer to (13) as the Aug-PI-CMO dynamics.
- If $K_p = 0$, (13) is equivalent to Aug-PDGD.

Convergence analysis

Theorem (Global exponential convergence)

Let us assume that C is full rank and there exist $0 < \underline{c} \leq \bar{c}$ such that $\underline{c}I \preceq CC^\top \preceq \bar{c}I$. Let $\rho < \bar{c}^{-1}$. Then, there exist real positive constants α_1 and α_2 such that

$$\|x(t) - x^*\|_2 \leq \alpha_1 e^{-\frac{1}{2}\mu t}, \quad \|\lambda(t) - \lambda^*\|_2 \leq \alpha_2 e^{-\frac{1}{2}\mu t} \quad (14)$$

where

$$\mu \leq \min \left\{ \frac{1}{2} K_p \underline{c}, \frac{2K_i \underline{g} - K_p \bar{g}^2}{K_i} \right\} \quad (15)$$

where $0 < \underline{g} \leq \bar{g}$ are assessed in the proof.

Convergence analysis

Proof (Sketch). Let $z^* = [x^{*\top}, \lambda^{*\top}]^\top$ and $z(t) = [x^\top(t), \lambda^\top(t)]^\top$.

We check that

$$V(z(t)) = (z(t) - z^*)^\top \begin{pmatrix} \sigma I_n & 0 \\ 0 & I_m \end{pmatrix} (z(t) - z^*) \quad (16)$$

where $\sigma > 0$, satisfies

$$\dot{V}(z(t)) \leq -\mu V(z(t)). \quad (17)$$

Then, by the Lyapunov Theorem, the result follows. $\square$

In the proof, we assume $K_i = \sigma$ to simplify the computations. Other choices may enhance the convergence rate.

Comparison with Aug-PDGD

- The proof of Theorem 6 is more straightforward than the proof of Theorem 2 in [Qu and Li, 2019] because the diagonal form of the Lyapunov function (17) reduces the computations when compared to the Lyapunov function with cross terms in [Qu and Li, 2019].
- A theoretical comparison of the convergence rates μ and τ in [Qu and Li, 2019] is challenging due to the fact τ depends on many constants that are not easy to assess. Instead, we can estimate μ straightforwardly.

We consider the simple problem:

$$\min_{x \in \mathbb{R}} \frac{1}{2}wx^2 \quad s.t. \ x \leq 0, \quad \text{with } w > 0. \quad (18)$$

In this case, both Aug-PDGD and Aug-PI-CMO correspond to a switched LTI system with two modes.

- ① In the first mode, they enjoy the same convergence rate.
- ② In the second mode, for PDGD, it is impossible to increase the convergence rate beyond $w + \rho$. Conversely, for PI, the eigenvalues are complex conjugate by choosing K_i sufficiently large, and the convergence rate is $K_p + w + \rho$.

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Example 1: Quadratic programming

$$\begin{aligned} x^* &= \arg \min_{x \in \mathbb{R}^n} \frac{1}{2} x^\top \left(I + W^\top W \right) x + b^\top x \\ \text{s.t.} \\ Cx - d &\leq 0 \end{aligned} \tag{19}$$

where $W \in \mathbb{R}^{n,n}$, $b \in \mathbb{R}^n$, $C \in \mathbb{R}^{m,n}$, $d \in \mathbb{R}^m$ are randomly generated vectors or matrices, with independent, normally distributed components.

- We set $n = 50$ and $m = 45$;
- We solve the optimization problem using the proposed algorithm (13) and Aug-PDGD. We integrate the differential equations in the interval $[0, 30]$ s through the *ode45* MATLAB command.
- We set $K_i = \eta = 1$ and $K_p = -0.7$.
- We perform 100 random runs with different realizations of W, b, C, d .

Example 1: Quadratic programming

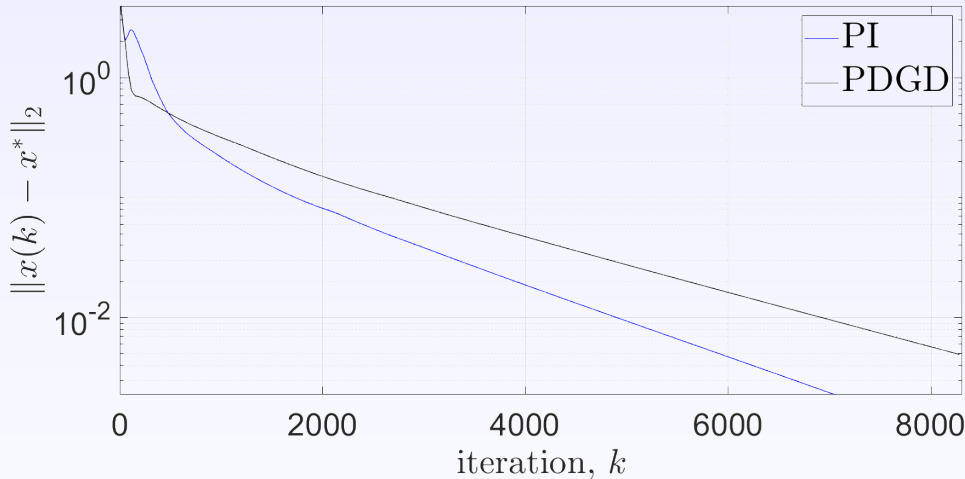


Figure: Example 1: distance from the optimum.

Example 1: Quadratic programming

PI requires fewer iterations than PDGD in all the runs and, consequently, less computational time.

	mean	standard deviation	worst case
N Aug-PDGD	8128.3	491.9	9361
N Aug-PI-CMO	6903.2	312.1	7613
T Aug-PDGD	1.23×10^{-1}	8.04×10^{-3}	1.53×10^{-1}
T Aug-PI-CMO	1.02×10^{-1}	4.74×10^{-3}	1.15×10^{-1}

Table: Example 1: results over 100 random runs. N is the required numbers of iteration; T the computational time (in seconds).

Example 2: Linear system identification

We consider the problem of identifying an unknown stable linear dynamic system $H(z)$ using uncertain input-output measurements $\{u_k, \tilde{y}_k\}$, for $k = 1, \dots, N$:

$$\tilde{y}_k = H(q^{-1})u_k + \eta_k. \quad (20)$$

To perform the identification, we select a linear-in-the-parameters model structure $\tilde{H}(z, \theta)$ of the form

$$\tilde{H}(z) = \sum_{i=1}^P \theta_i B_i(z). \quad (21)$$

where, for simplicity,

$$B_i(z) = \frac{z}{z - p_i}, \quad p_i \in \{-0.9, -0.85, \dots, 0.9\}. \quad (22)$$

Example 2: Linear system identification

We look for the value of the parameter θ that minimizes the ℓ_∞ -norm of the simulation error

$$\theta^* = \arg \min_{\theta \in \mathbb{R}^P} \|y_k(\theta) - \tilde{y}_k\|_\infty. \quad (23)$$

By adding a slack variable $\Delta \in \mathbb{R}$, we recast problem (23) to the following linear programming problem:

$$\begin{aligned} \theta^*, \Delta^* &= \arg \min_{\theta \in \mathbb{R}^P, \Delta \in \mathbb{R}} \Delta \\ \text{s.t.} \\ -\Delta &\leq Z_k \theta - \tilde{y}_k \leq \Delta, \quad k \in \{1, \dots, N\} \end{aligned} \quad (24)$$

where $Z_k = [z_1(k), \dots, z_P(k)] \in \mathbb{R}^P$, $z_i(k) = B_i(q^{-1})u(k)$.

Example 2: Linear system identification

- We solve the optimization problem using the proposed PI algorithm in (13) and Aug-PDGD. We integrate the differential equations in the interval $[0, 1000]$ s through the *ode23* MATLAB command.
- We set $K_i = \nu = 1$ and $K_p = -0.5$.
- We obtain the same $\text{FIT} = 98.5\%$ for both the identified models.
- The PI algorithm is about two times faster than Aug-PDGD.

	N	T
Aug-PDGD	10^6	679.46
Aug-PI-CMO	5×10^5	233.09

Table: Example 2: N is the required numbers of iteration; T the computational time (in seconds).

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Conclusions

- By elaborating on the CMO approach proposed in [Cerone et al., 2024], we extend it to inequality-constrained problems.
- Based on PI control, we develop a variant of Aug-PDGD in which an additional term adjusts the dynamics of the Lagrange multipliers.
- We prove global exponential convergence and show that the proposed approach enhances the convergence speed.
- Ongoing follow-up work envisages the relaxation of affinity and smoothness requirements in the considered model.

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-  Cerone, V., Fosson, S. M., Pirrera, S., and Regruto, D. (2024).
A new framework for constrained optimization via feedback control of Lagrange multipliers.
[Arxiv preprint arXiv:2403.12738. Submitted to IEEE Trans. Autom. Control.](#)
-  Qu, G. and Li, N. (2019).
On the exponential stability of primal-dual gradient dynamics.
[IEEE Control Syst. Lett.](#), 3(1):43–48.